LET Science: Inquiry, Systems, and Evidence-Based Explanations

LET · science · BoardPrep lesson v1

Official examination scope

DepEd's K–12 Science curriculum emphasizes scientific knowledge, process skills, and evidence-based explanations across life science, chemistry, physics, and Earth science. This lesson covers selected competencies and is not a substitute for the current PRC TOS or full curriculum guide.

Learning objectives

- Distinguish an observation from an inference and a testable question.
- Explain diffusion using particle movement and a concentration gradient.
- Use a model and evidence to interpret motion or an Earth/life-science pattern.

1. Inquiry, variables, and evidence

A scientific explanation connects observations to a model and makes clear what evidence could support or challenge it. In an investigation, define the question, independent variable, measured outcome, and conditions held comparable. Repeat measurements where appropriate and record uncertainty rather than hiding variation. A conclusion should follow from the data and design; a single uncontrolled observation rarely establishes cause. Models are useful representations, not literal copies of nature, and their limits should be stated when applying them to a new situation.

2. Cells, particles, and movement across membranes

Particles in matter are in continuous motion. Diffusion is the net movement of particles from a region of higher concentration toward lower concentration; it does not require a cell to spend energy for that net movement. Osmosis refers specifically to water moving across a selectively permeable membrane under a concentration difference. Active transport uses cellular energy to move substances against a gradient. A diagram should be read for what is moving, which direction it moves, and whether energy is shown before choosing a process name.

3. Motion, forces, and energy models

Position change over time describes motion; a distance–time or position–time graph shows how that change varies. The slope of a position–time graph represents velocity under the graph's chosen axes. A net force changes an object's motion, while balanced forces can coexist with motion at constant velocity. Energy accounting tracks transfers and transformations without treating energy as destroyed in an ideal closed model. Check sign, unit, direction, and the conditions of a model before interpreting a graph or calculation.

4. Earth, ecosystems, and environmental reasoning

Earth and life-science patterns reflect interacting systems. Relative rock-layer order can support an age sequence where the layers have not been disturbed, but it does not by itself provide a calendar date. In an ecosystem, energy flows through trophic relationships while matter cycles; population patterns depend on resources, interactions, and environmental conditions. Environmental claims should identify the measured outcome, spatial scale, and time period, then distinguish evidence from a proposed cause. Connect local examples to the curriculum without assuming one community represents all settings.

Identify movement across a membrane

Question: Particles move from higher concentration to lower concentration across a selectively permeable membrane, with no energy input. Which process is illustrated?

1. Identify the substance as particles; the prompt does not specify water alone.
2. Note the net direction is down the concentration gradient.
3. Check that no energy input is described.
Answer: Simple diffusion.
Check: If the moving substance were specifically water, osmosis would be the more specific term; movement against a gradient would require another explanation.

Common misconceptions

- Every movement across a membrane is osmosis.
- A model or graph is a literal record of all conditions.
- An observed association alone identifies the cause.

Practice patterns

- Name the observation before the interpretation.
- Read graph axes and units before inferring change.
- State one limitation of the experiment or model.

Official scope references

Official scope: Enhanced Tables of Specifications for the Licensure Examination for Professional Teachers — Professional Regulation Commission; Professional Regulatory Board for Professional Teachers — https://prc.gov.ph/Pages/PRBv4/ProfessionalTeachersv9.htm
Official scope: PRC Guidelines on the Implementation of the PRC-CHED JMC on LEPT Alignment — Professional Regulation Commission — https://www.prc.gov.ph/article/guidelines-implementation-prc-ched-jmc-alignment-teacher-education-curriculum

Instructional references

Instructional reference: DepEd K to 12 Curriculum Guide: Science — Department of Education, Philippines — https://lrmds.deped.gov.ph/detail/30/5459
Instructional reference: Biology 2e — OpenStax, Rice University — https://openstax.org/details/books/biology-2e
Instructional reference: Chemistry 2e — OpenStax, Rice University — https://openstax.org/details/books/chemistry-2e
Instructional reference: University Physics Volume 1 — OpenStax, Rice University — https://openstax.org/details/books/university-physics-volume-1

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